



Nitigya Chandel

Roll No.: 2023ug4010

B. Tech

ECE (with ES and IoT)

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EDUCATION

Indian Institute of Information Technology Ranchi

2027

B.Tech in Electronics and Communication Engineering (Specialization in ES and IoT)

CGPA: 8.06 / 10.0

TECHNICAL SKILLS

Languages: C/C++, JavaScript, TypeScript, Python, Java, SQL, HTML/CSS

Developer Tools: Linux, Git/GitHub, Vercel, Render, Arduino

Frameworks: React.js, Node.js, Express.js, Prisma, Supabase, Redux Toolkit, Tailwind CSS

Cloud/Databases: MongoDB (NoSQL), MySQL (SQL), PostgreSQL (via Supabase/Prisma)

Coursework: Data Structures & Algorithms, Operating Systems, Object-Oriented Programming, Embedded Systems / Microcontrollers

PERSONAL PROJECTS

Nectar 🌱 🌱

March 2026

Tools: TypeScript, PostgreSQL (via Prisma), Supabase, React, RTK, Gemini API

- Developed a **type-safe, relational backend architecture** utilizing **PostgreSQL** and **Prisma** to securely manage complex user biometrics and strict dietary constraints.
- Leveraged the **Gemini API** to dynamically generate highly personalized, **macro-accurate 5-meal daily plans**, processing physical attributes against targeted physiological goals.
- Orchestrated robust global state management via **Redux Toolkit (RTK)** to manage complex client-side data flows, ensuring seamless synchronization of user biometrics, macro tracking metrics, and dynamic UI states.

GameVault 🌱 🌱

Jan 2026

Tools: React.js, MongoDB, Mongoose, JWT, Bcrypt, Framer Motion, RAWG API

- Formulated a custom **hardware benchmarking algorithm** that cross-references user-defined system specifications against queried **RAWG API** data to dynamically validate minimum and recommended hardware compatibility.
- Designed a **scalable user-library database schema** utilizing **MongoDB** and **Mongoose**, allowing users to persistently categorize complex game collections across discrete tracking states (Played, On-Hold, Dropped, Want to Play).
- Incorporated the **RESTful RAWG API** to dynamically populate the application with expansive video game meta-data, handling asynchronous data fetching and client-side rendering with **React.js**.
- Established **secure user authentication** and authorization pipelines utilizing **JWT** for session integrity and **bcrypt** for cryptographic password hashing to protect personal user libraries.
- Built a highly responsive user interface featuring **Framer Motion physics**, a global search utility, and a personalized rating engine that dynamically maps UI state changes to user sentiment scores.

HydraFS: Shard-Based Distributed Storage System 🌱

April 2026

Tools: C++17, Boost.Asio, Protocol Buffers, TCP/IP, Raspberry Pi

- Architected a high-performance distributed filesystem in **C++17** utilizing **Boost.Asio** for non-blocking TCP networking, reliably slicing and transmitting large binary payloads as discrete 1MB shards.
- Integrated **Google Protocol Buffers (Protobuf)** for fast, language-agnostic data serialization and engineered an asynchronous **Switchboard** system to seamlessly route telemetry heartbeats and data packets in real-time.
- Deployed a fault-tolerant storage vault on a **Raspberry Pi**, implementing half-close TCP stream flushing and ensuring 100% data fidelity through post-transfer **MD5 checksum** verification.

ACHIEVEMENTS

CodeChef 3 Star Rating (Max Rating: 1727)

May 2024

- Achieved a **peak global rating of 1727**, demonstrating advanced proficiency in **data structures and algorithmic problem-solving**.

Codeforces Pupil (Max Rating: 1355)

May 2025

- Attained the **Pupil rank** by consistently participating in competitive programming contests and solving **complex logic constraints** under time pressure.